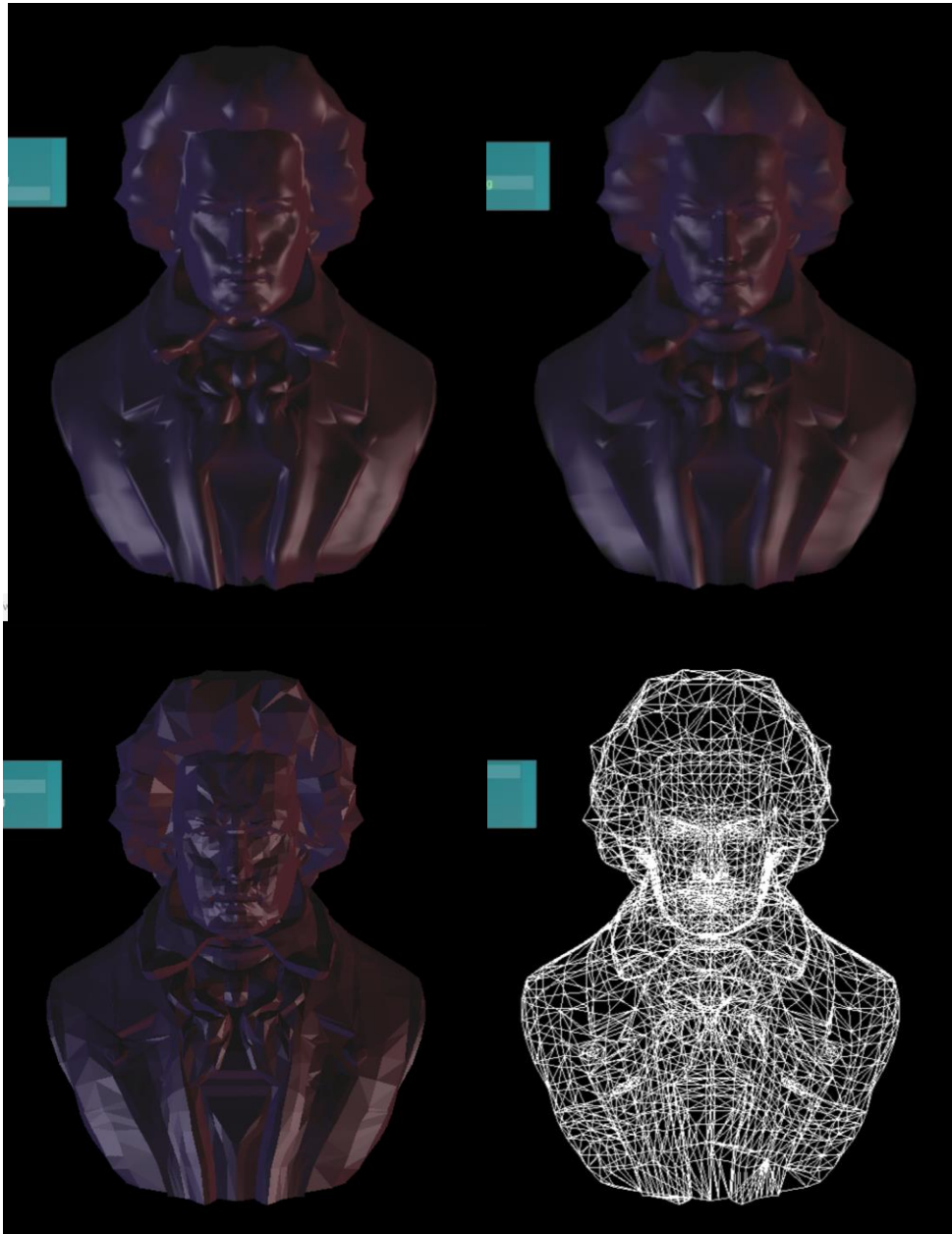


Computer Graphics – Assignment 3

Demonstration Paper



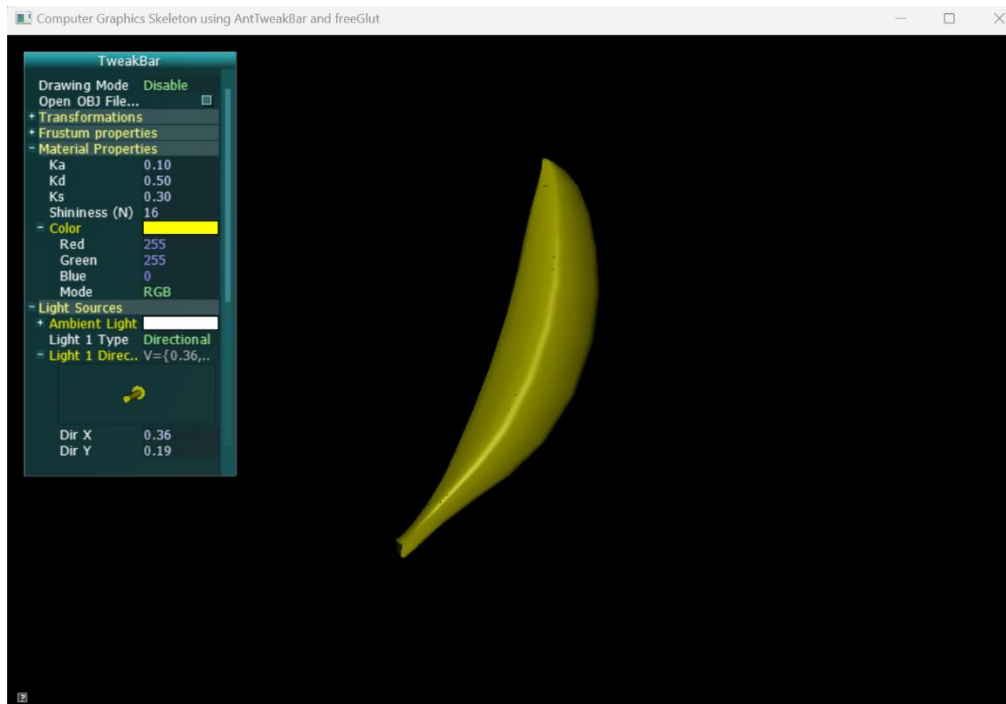
Shahar yuval – 214859589

Date of submission – 01/12/2025

Note: of course that all previous features are still compatible (transformations, screen adjustments, normal, coordinates, BB, lookAt and more)

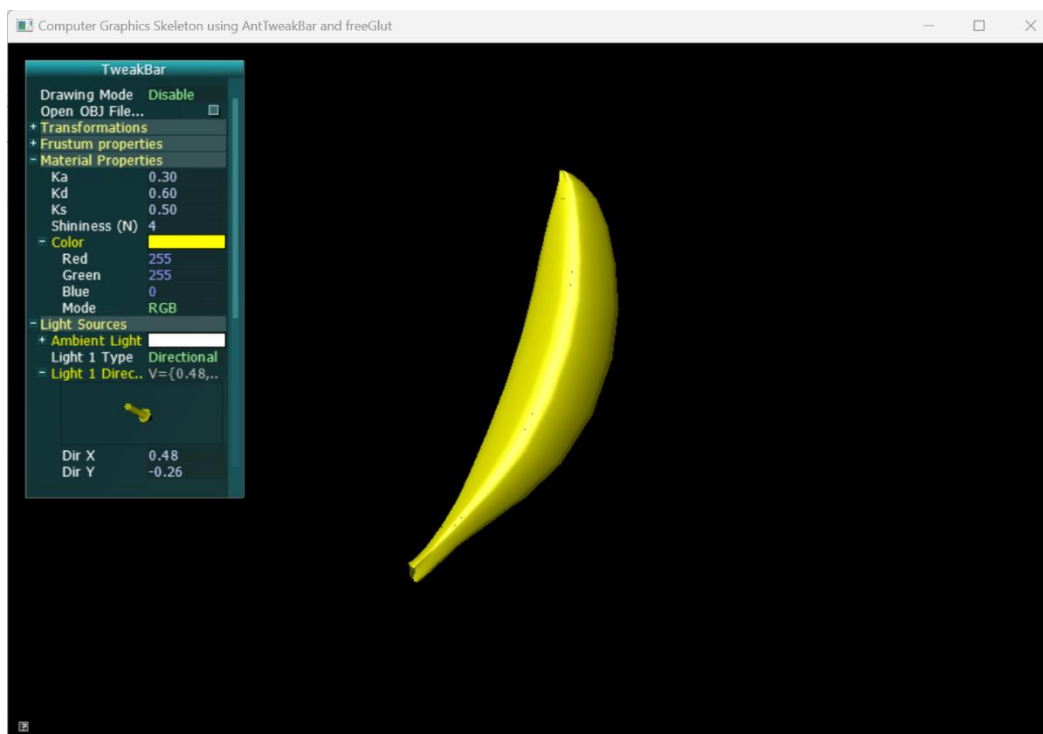
Part 1 – Material

Object color:



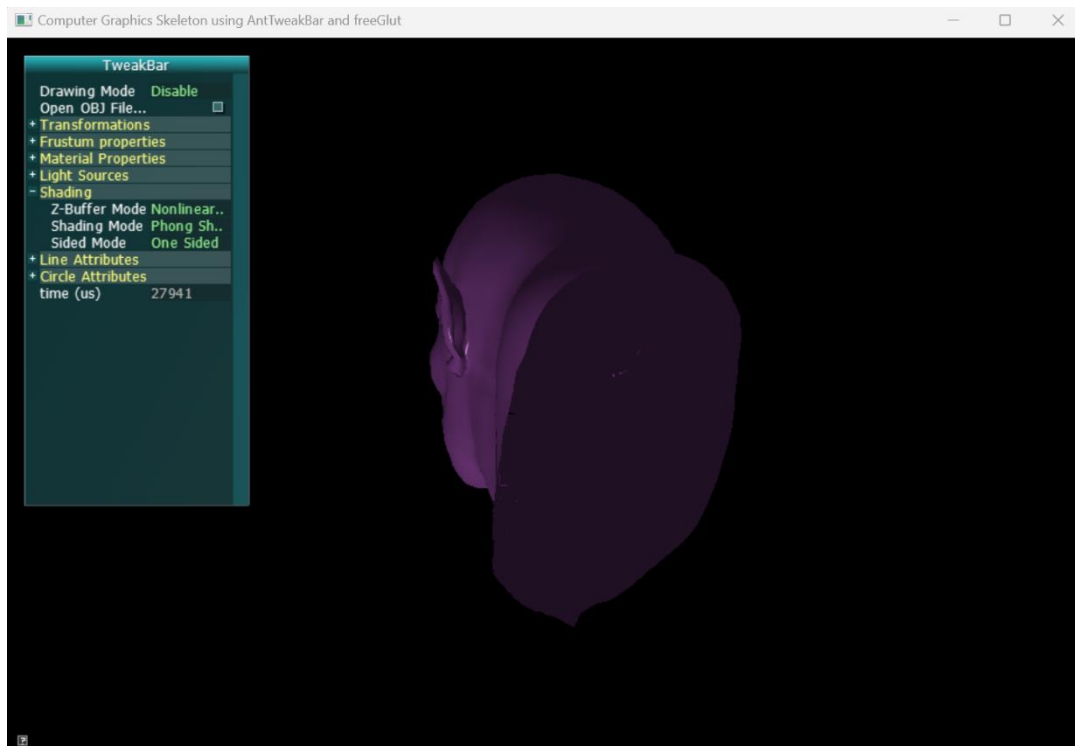
Rendered banana.obj with changed material color

Ambient, diffuse and spectral constants:



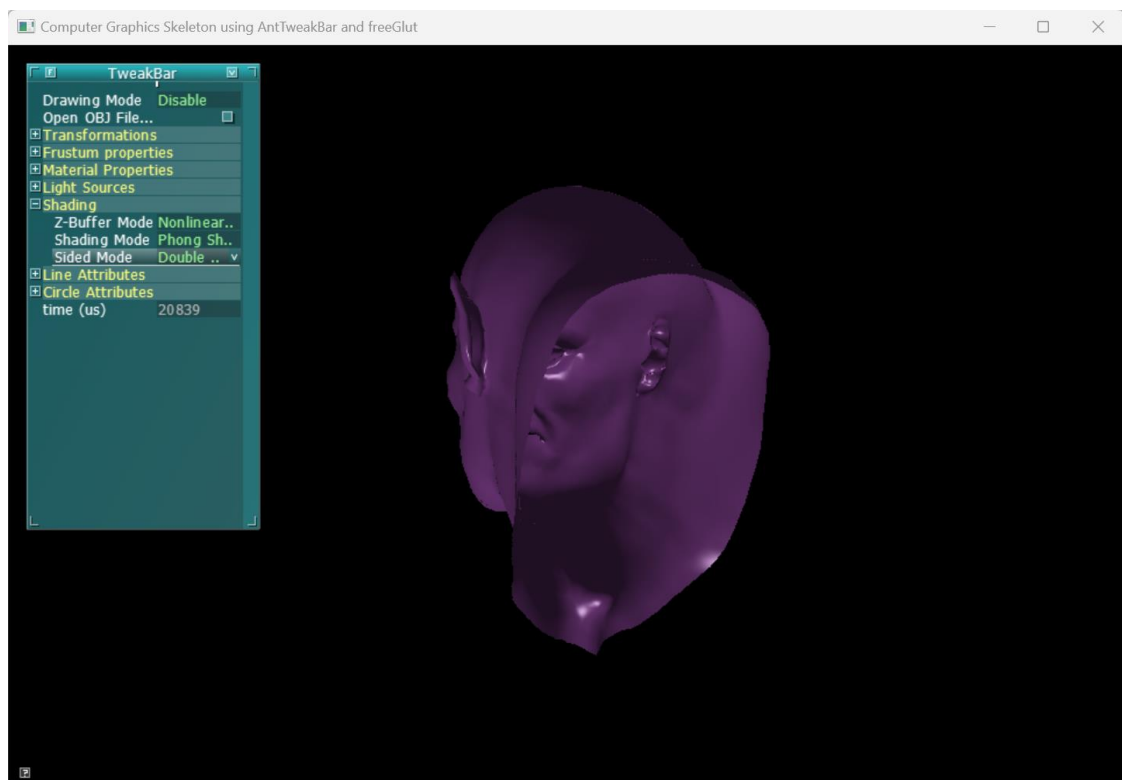
Render of the same object and material color, changed k_a , k_d , k_s , n

Single sided / double sided:



Render of beast head model. On **one sided** mode

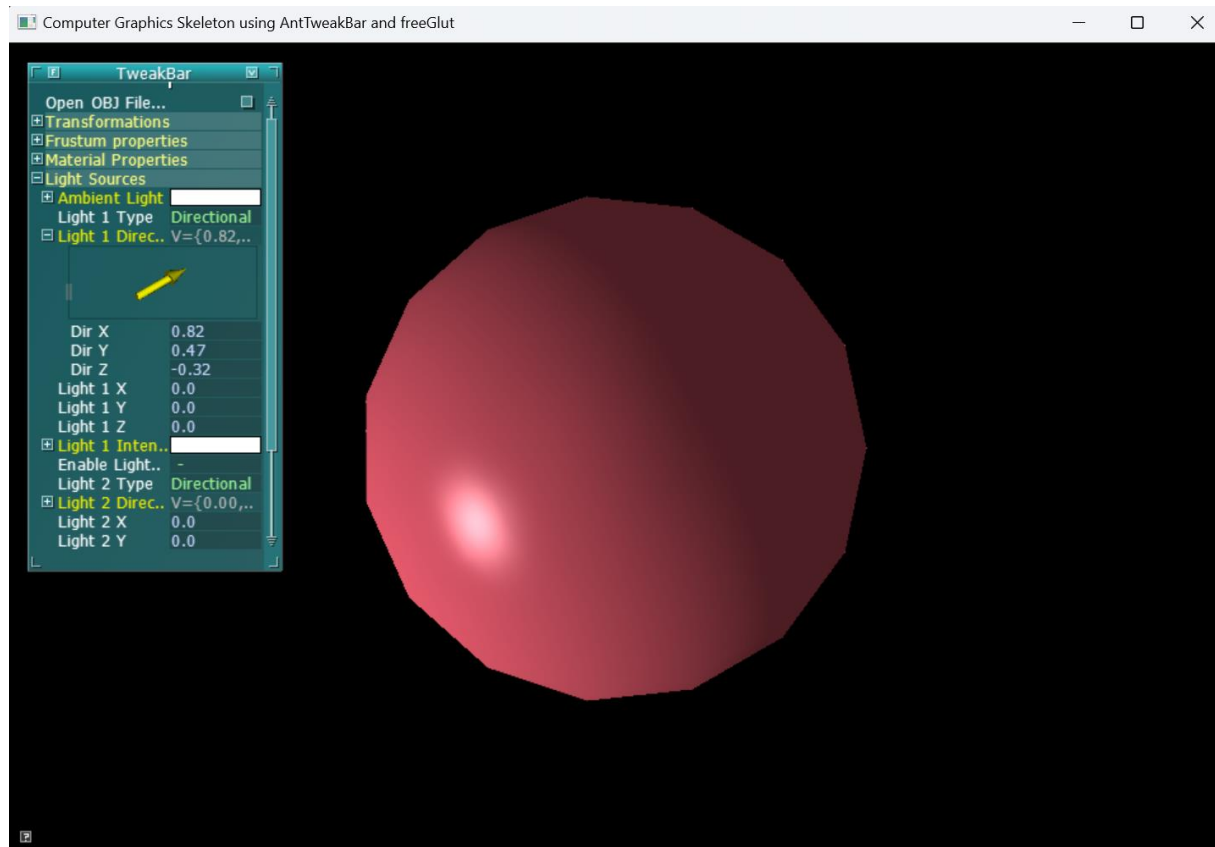
(note: back facing surfaces are only lit by ambient light)



Render of beast head model. On **double sided** mode

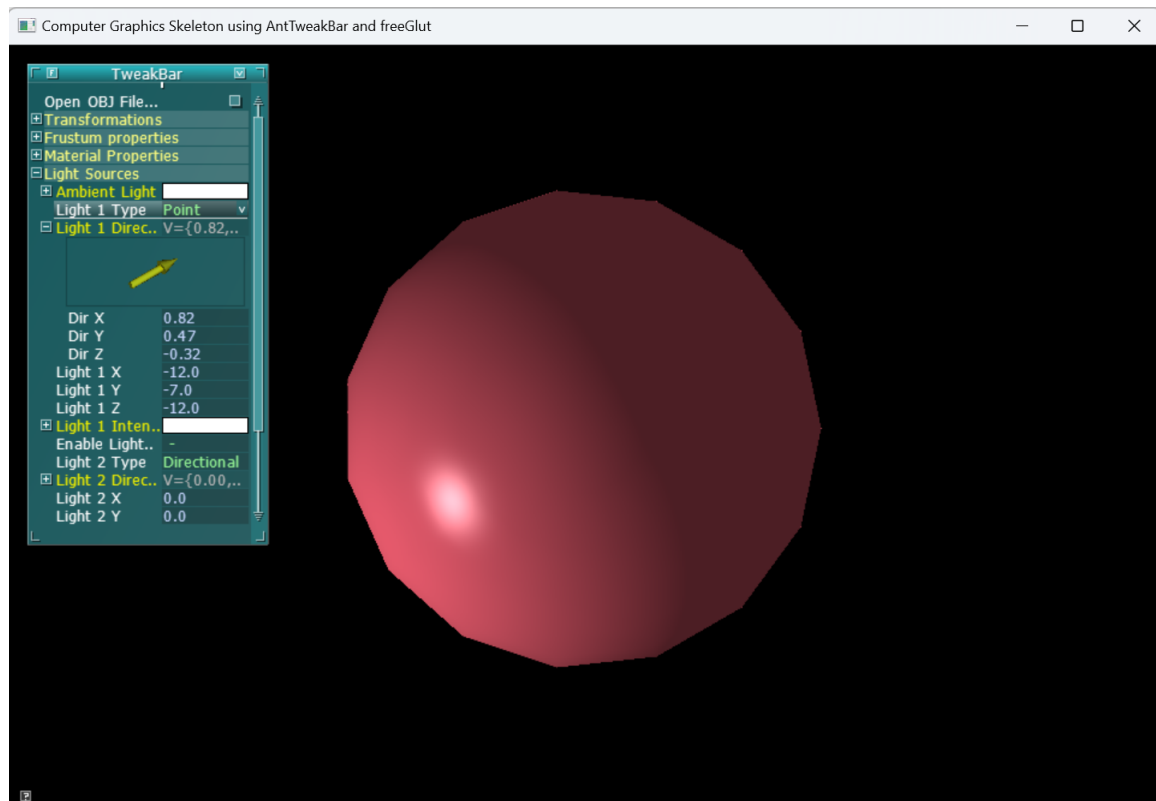
Part 2 – Lighting

Directional / Point and 1 / 2 light sources:



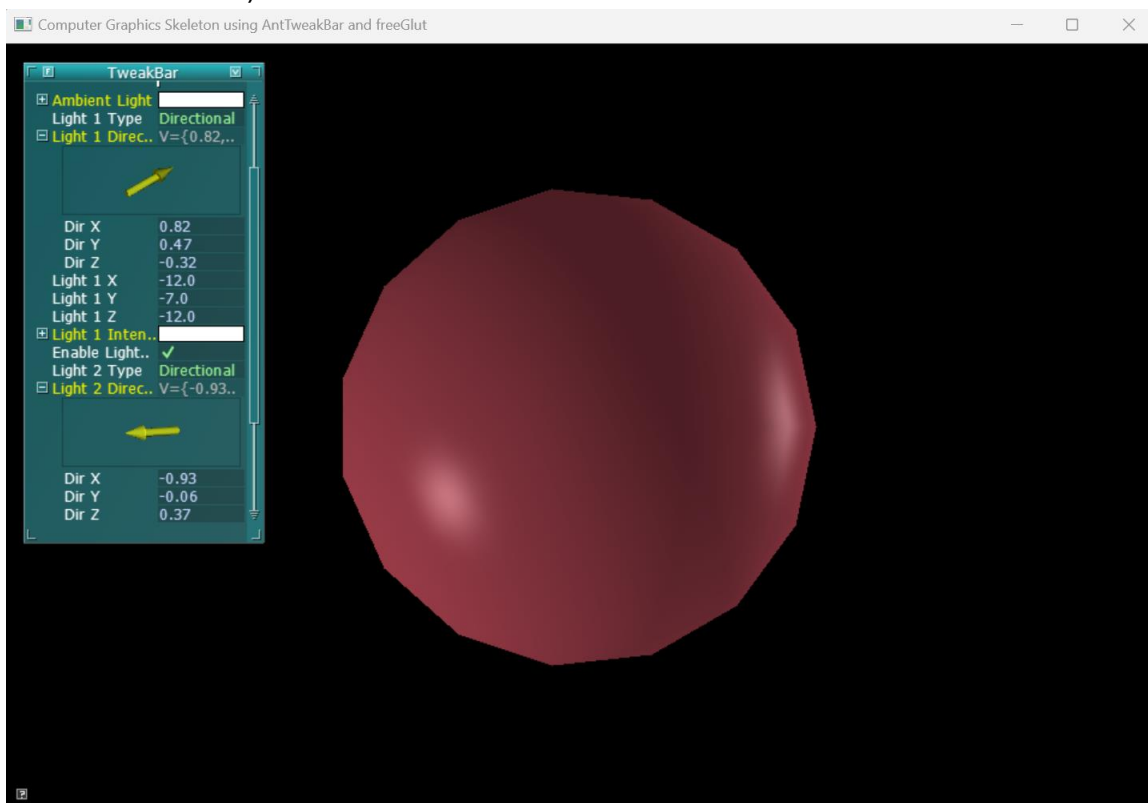
Render of sphere model. With 1 Directional light source

(note: direction can be seen in tweak bar)



Render of sphere model. With 1 light Point light source

note: I tried that the light source would come from the same direction as before, and still there are noticeable differences, (most noticeable difference is that the spectral spot seems smaller)

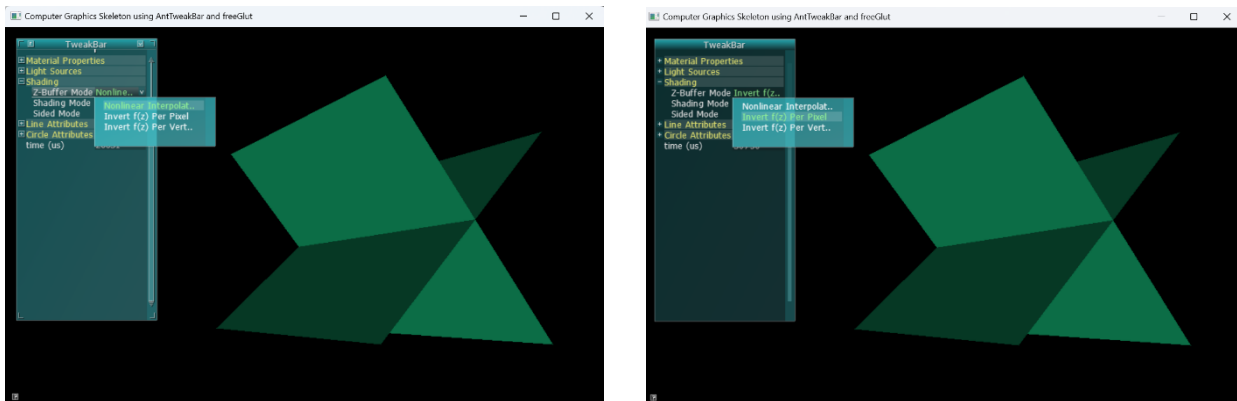


Render of sphere model. With 2 directional light sources

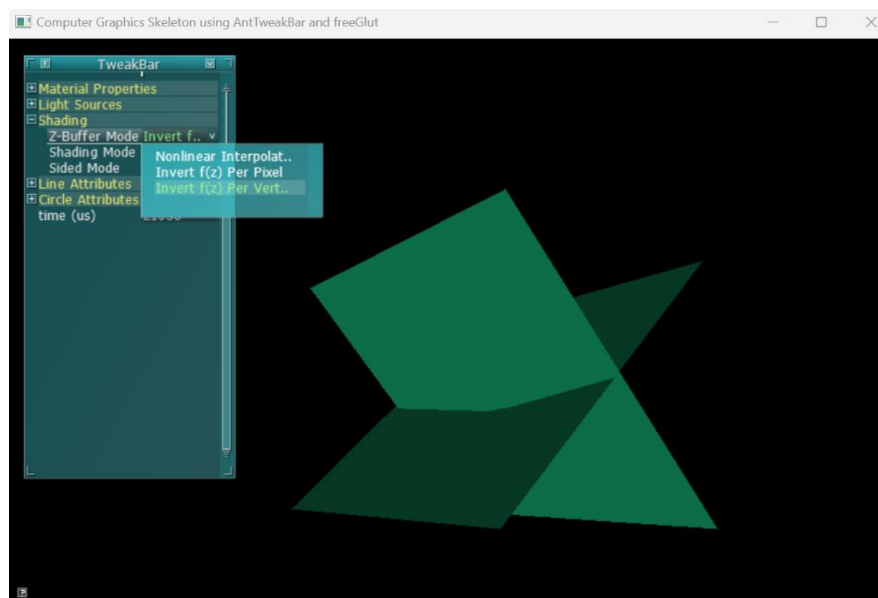
Note: the program also has these features

- Ambient light
- freedom in light color for both lights and for ambient light
- light color is clamped in overflow cases

Part 3 – the Renderer



Mode 1- on the left, Mode 2- On the right, Mode 3- one the Bottom



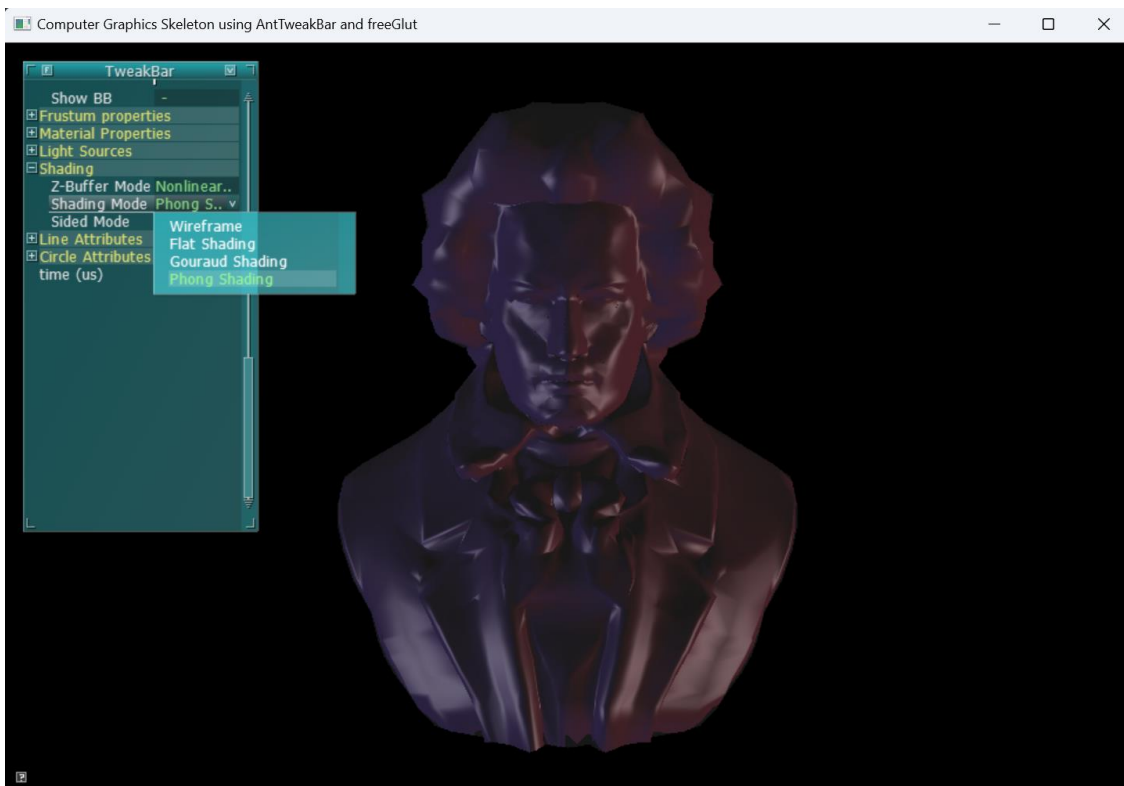
Both mode 1 and 2 are supposed to give good results (because depth is linear in $f(z)$ in perspective space) and this is what we got.

Moreover, we don't see difference between Mode 1 and Mode 2. This is because they both are supposed to give the same result (the correct result) but their difference lies in the computational instability of Mode 2 (because the $f^{-1}(f(z))$ calculation is unstable) but apparently this instability is unnoticeable in our case.

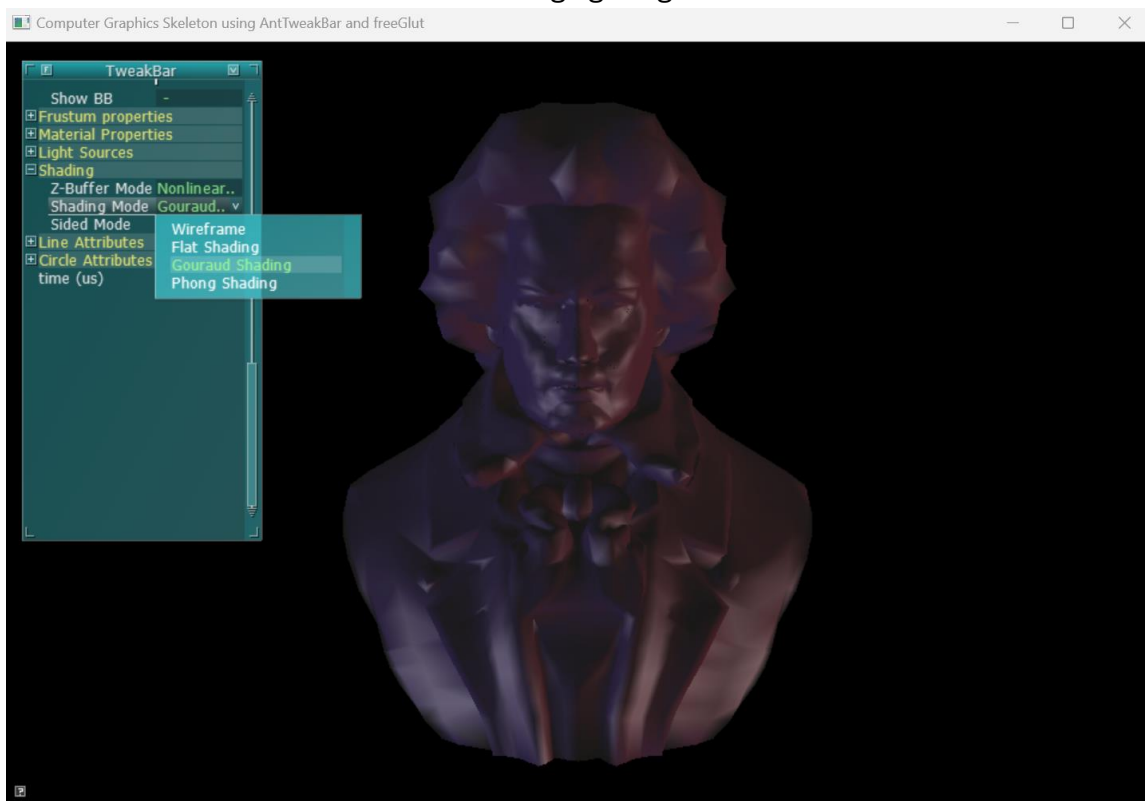
In Mode 3 we can see that we don't get the correct result but we get a “curved intersection” this is because z is not linear in the perspective space.

Shading modes:

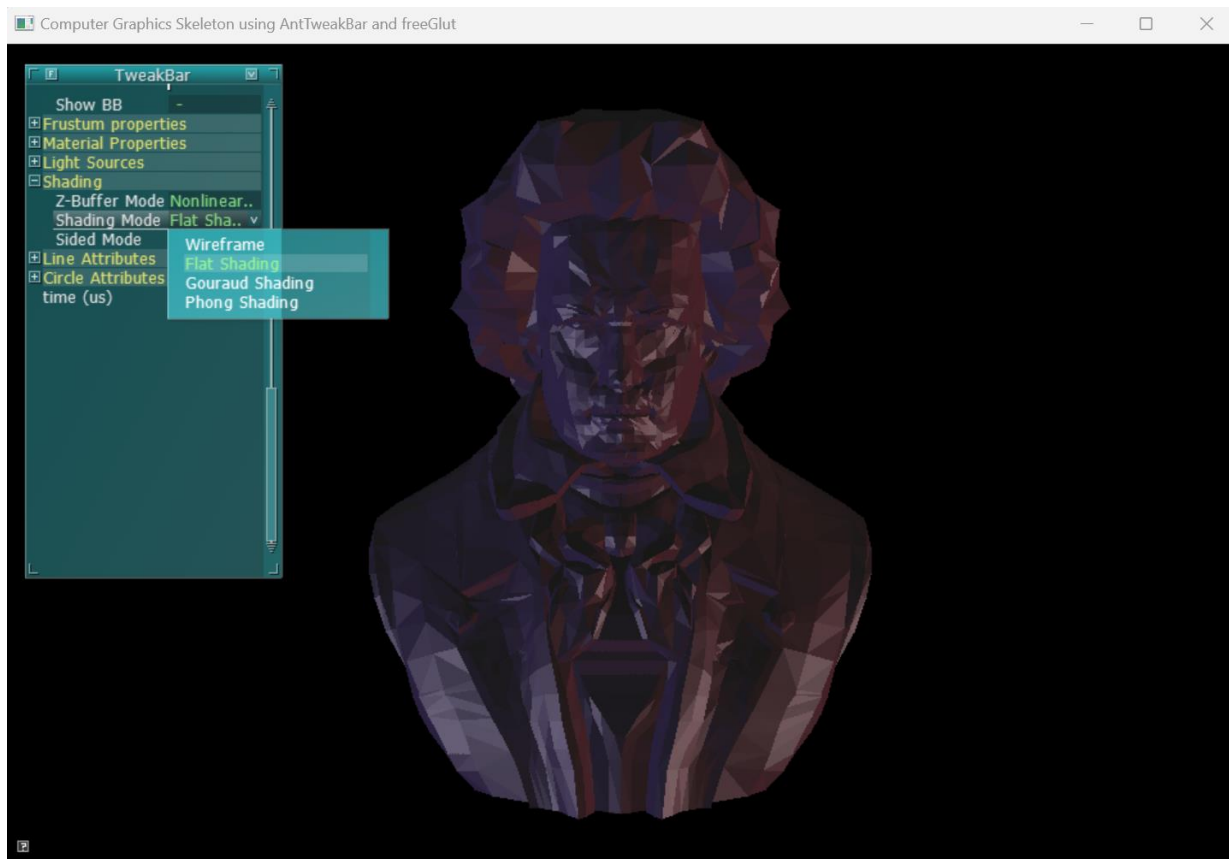
Note: all pictures are renders of Beethoven model illuminated by a blue light source from the right and a red light source from the left



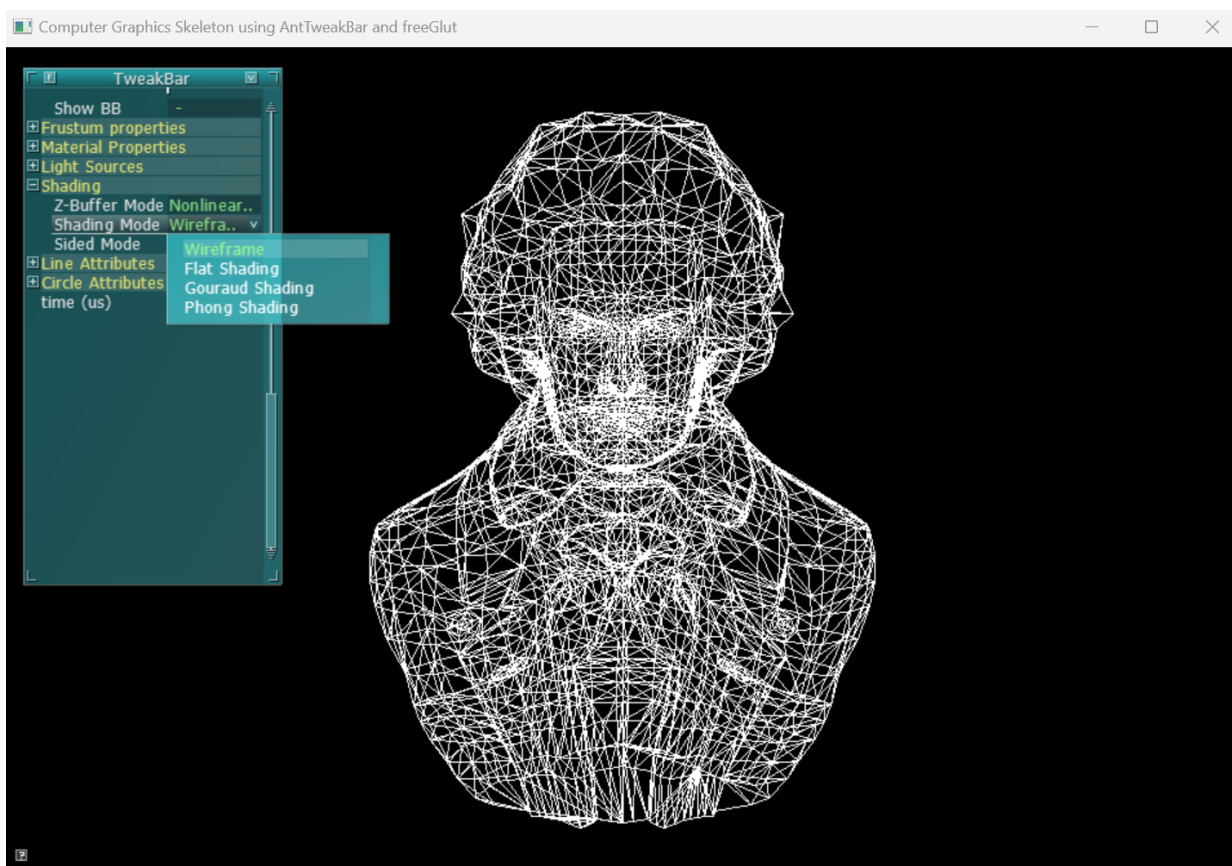
Phong lighting



Gouraud Shading



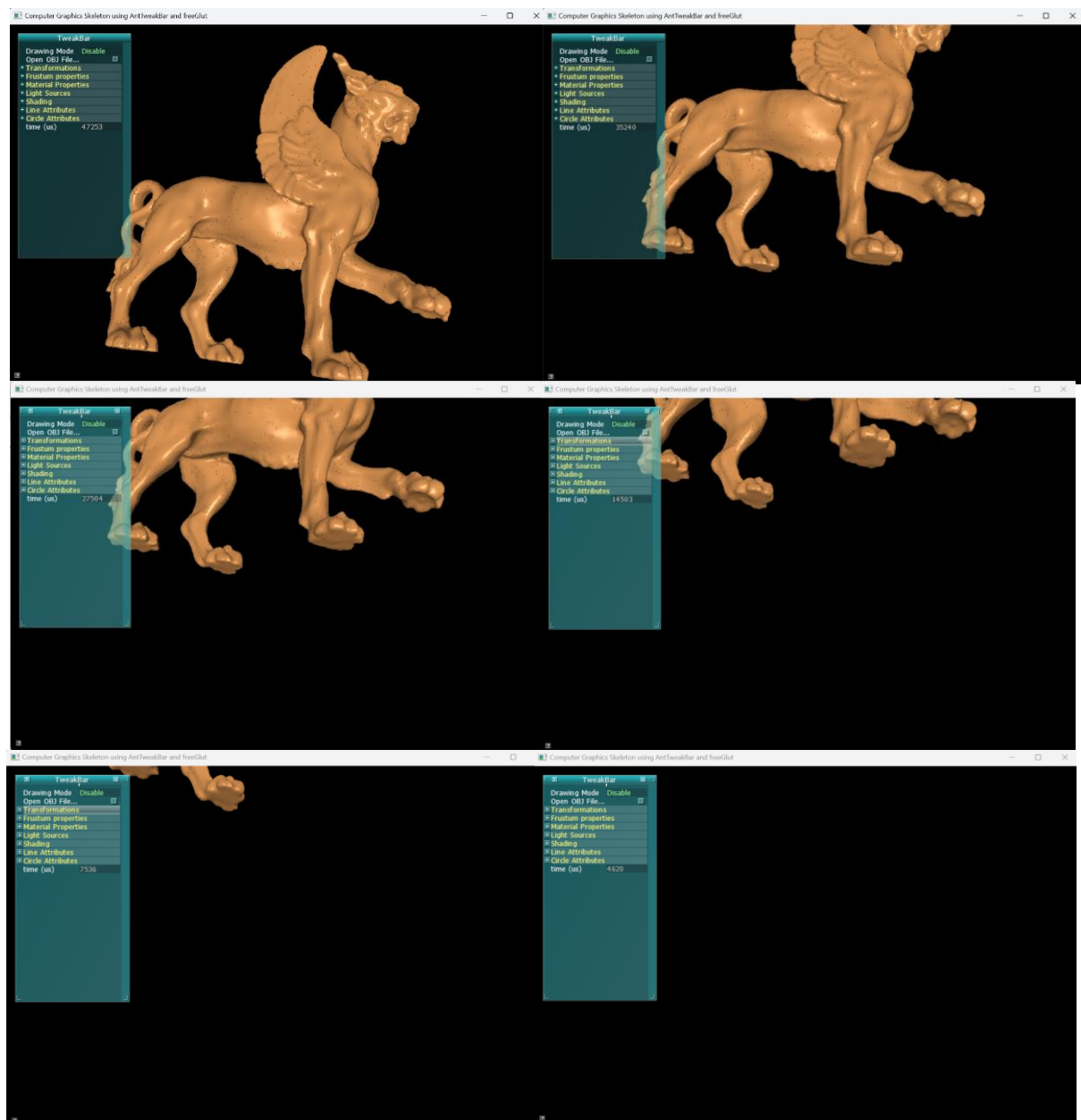
Flat shading



Wireframe rendering

Clipping

The program performs clipping. This will be shown by seeing the increased performance when the object partially exits the screen:



The time values are :

- image 1 – 47,253[μ s]
- image 2 – 35,256[μ s]
- image 3 – 27,504[μ s]
- image 4 – 14,503[μ s]
- image 5 – 7,536[μ s]
- image 6 – 4,620[μ s]

descending, which validates the use of clipping.